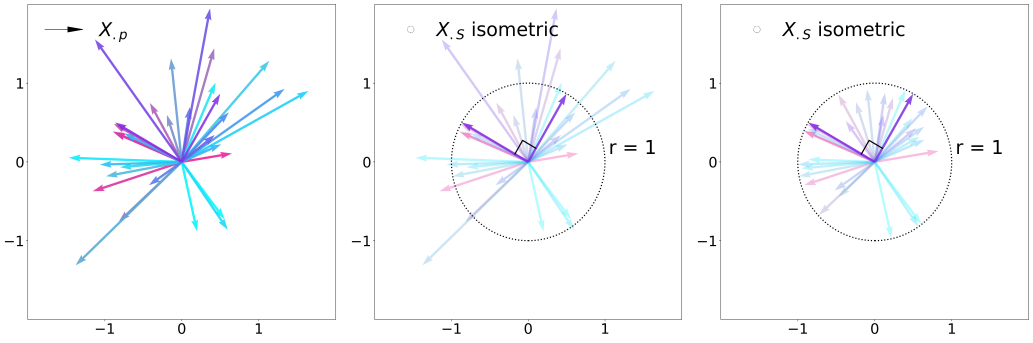


(a) Raw data (b) Isometric subset (c) Normalization step

Figure 1: 1a Raw data X represented as column-vectors. 1b An isometric subset - the identification of which is our objective. 1c The vectors after normalization so that vectors of length 1 maintain longest length.

Isometry Pursuit: Convex Recovery of Orthonormal Column Submatrices

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Abstract

Isometry pursuit is a convex algorithm for identifying orthonormal column-submatrices of wide matrices. It consists of a vector normalization followed by multitask basis pursuit. Applied to Jacobians of putative coordinate functions, it helps identity locally isometric embeddings from within interpretable dictionaries. We provide theoretical and experimental results justifying this method, including a proof with realistic assumptions that such isometric submatrices, should they exist, are contained within the obtained support. For problems involving coordinate selection and diversification, it offers a synergistic alternative to greedy and brute force search.

1 Introduction

Many real-world problems may be abstracted as selecting a subset of the columns of a matrix representing stochastic observations or analytically exact data. This paper focuses on a simple such problem. Given a rank D matrix $X \in \mathbb{R}^{D\times P}$ with $P > D$, select a square submatrix $X_{\cdot S}$ where subset $S \subset [P]$ satisfies $|S| = D$ that is as isometric as possible.

This problem arises in interpretable learning specifically because while the coordinate functions of a given feature space may have no intrinsic meaning, it is sometimes possible to generate a dictionary of interpretable features for consideration as potential parametrizing coordinates. When this is the case, selection of candidate interpretable features as coordinates can take the above form. While implementations vary across data and algorithmic domains, identification of such coordinates generally aids mechanistic understanding, generative control, and statistical efficiency.

This paper shows that an adapted version of the algorithm in Koelle et al. (2024) leads to a convex alternative to the greedy approaches such as those in **??**Kohli et al. (2021); **?** for finding isometries. This algorithm - IsometryPursuit - is multitask basis pursuit applied to an appropriately normalized X selects orthonormal submatrices. Given vectors in $\mathbb{R}^D$,

the normalization log-symmetrizes length and favors those closer to unit length, while basis pursuit favors those which are orthogonal. Our theoretical results formalize this intuition, and show the usefulness of IsometryPursuit as a trimming procedure prior to brute force search for diversification and interpretable coordinate selection.

2 Background

Our algorithm is motivated by a classic question in numerical linear algebra.

2.1 Problem

Our goal is, given a matrix $X \in \mathbb{R}^{D \times P}$, to select a subset $S \subset [P]$ with $|S| = D$ such that $X_{.S}$ is as orthonormal as possible in a computationally efficient way. To this end, we define a ground truth loss function that measures orthonormalness, and then introduce a surrogate loss function that convexifies the problem so that it may be efficiently solved. Figure 1a and 1b provide a visual depiction of this problem when $D = 2$ and $P = 30$.

2.2 Isometry

The mathematical construct central to our estimation problem is the isometric linear map.

Definition 1 *A matrix $X \in \mathbb{R}^{D \times P}$ is an **isometry** or, equivalently, **unitary** if*

$$X^T X = I_D. \tag{1}$$

Note that necessarily, $P \geq D$.

Definition 1 has several equivalent formulations.

Proposition 1 *The singular values $\sigma_1(X) \dots \sigma_D(X)$ are equal to 1 if and only if X is an isometry.*

On the other hand, if $P = D$, X

Proposition 2 *The component vectors $X_{.}1 \dots X_{.D} \in \mathbb{R}^D$ form an orthonormal matrix if and only if, for all $d_1, d_2 \in [D], \langle X_{.d_1}, X_{.d_2} \rangle = \begin{cases} 1 & d_1 = d_2 \\ 0 & d_1 \neq d_2 \end{cases}$.*

That is, X is an isometry if its columns are orthonormal.

The applications of isometry are manifold. Overall, is a classically interesting problem. It is naturally a type of diversification. Embedding algorithms like Local Tangent Space Alignment (**?**), Multidimensional Scaling (**?**) and Isomap (**?**) non-parametrically estimate embeddings that are as isometric as possible. Note that an isometric embedding is a map whose differential induces isometry between tangent spaces. The inspiration for this paper comes from an approach that stitches together isometries selected from a dictionary to form global embeddings (Kohli et al., 2021). This method constructs such isometries through greedy search, with putative dictionary features added one at a time.

2.3 Functions minimized by isometries

The distortion loss function used the evaluate isometricness in (Kohli et al., 2021) reduces in the linear tangent space limit to the condition number. The *condition number* (**?**Boyd and Vandenberghe, 2004) $\frac{\sigma_1(X)}{\sigma_D(X)}$ measures the ratio of the largest and smallest singular values of a matrix A. While it is minimized by an isometric function, it is also minimized by any conformal (i.e. low distortion) linear map.

A class of functions more suitable for measuring isometry are S-divergences (**?**). S-divergences define a distance $D(P, P')$ on positive definite matrices P, P' given by

$$D(P, P') = (\log\det(\frac{P + P'}{2}) - \frac{1}{2}\log\det(PP'))^{\frac{1}{2}}. \tag{2}$$

Applied to the positive definite matrices X^TX and I_D, the S-divergence is a log singular value penalty.

Proposition 3 *If X^TX is positive definite, then*

$$D(X^TX, I_D)^2 = \sum_{d=1}^{D} \log \frac{\sigma_d(X) + \sigma_d(X)^{-1}}{2}. \tag{3}$$

By Proposition 1, this function is uniquely minimized by isometric matrices.

2.4 Best subset selection

Equipped with function $\ell(X)$ such as $D(X, I_D)$ that is minimized by an isometry, we now consider algorithms for solving problems of the form

$$\arg \min_{S \in \binom{[P]}{D}} \ell(X_{\cdot S}) \tag{4}$$

where $\binom{[P]}{D} = \{A \subseteq [P] : |A| = D\}$. Brute force algorithms consider all possible solutions. These algorithms are conceptually simple, but have the often prohibitive time complexity $O(C_\ell P^D)$ where C_ℓ is the cost of evaluating ℓ. Greedy algorithms consist of iteratively adding one element at a time to S. This algorithms have time complexity $O(C_\ell PD)$ and so are computationally more efficient than brute force algorithms, but can get stuck in local minima. Formal definitions are given in Section 4.1.

Sometimes, it is possible to introduce an objective which convexifies problems of the above form (**???**). Solutions

$$\arg\min f(\beta) : Y = X\beta \tag{5}$$

to the overcomplete regression problem $Y = X\beta$ are a classic example (**?**). When $f(\beta) = \|\beta\|_0$, this problem is non-convex, and is thus suitable for greedy or brute algorithms, but when $f(\beta) = \|\beta\|_1$, the problem is convex, and may be solved efficiently via interior-point methods. When the equality constraint is relaxed, Lagrangian duality may be used to reformulate as a so-called Lasso problem, which leads to an even richer set of optimization algorithms (**?**).

3 Method

The contributions of this paper is that we introduce an ℓ uniquely minimized by isometries that is

- convex in its parameters
- induces sparsity
- is more appropriate than S-divergence when an isometric submatrix is not present in that it always favors vectors of unit length

This ℓ is a multitask inspired by the group basis pursuit (Koelle et al., 2022; **?**) which we call differently since grouping is dependent only on the structure of matrix-valued response variable I_D. We apply the multitask basis pursuit paradigm used to select independent dictionary elements in Koelle et al. (2022; 2024) to the more specific problem of selecting pointwise isometries from a dictionary. We first define a ground truth objective computable via brute and greedy algorithms that is uniquely minimized by orthonormal matrices. We then define the combination of normalization and multitask basis pursuit that approximates this ground truth loss function. We finally give a brute post-processing method for ensuring that the solution is D sparse, and provide a theoretical result that the two stage approach will always result in recovery of an isometric solution from the dictionary, should one exist.

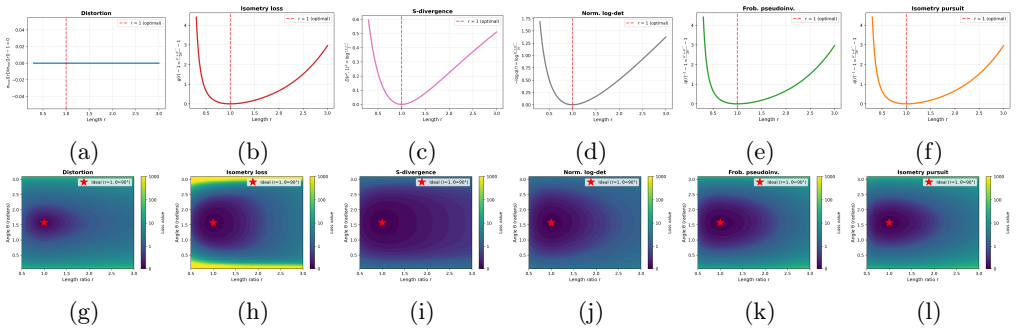


Figure 2: Comprehensive loss function comparison for isometry pursuit (2 rows × 6 columns). Top row: 1D loss curves for $X = [[r]]$ showing loss behavior versus vector length. Bottom row: 2D loss landscapes for 2×2 matrices parametrized by length ratio r and angle θ with shared colorbars. Red vertical dashed line in 1D plots marks $r = 1$ (optimal). Red star in 2D plots marks $(r = 1, \theta = \pi/2)$ (ideal). The panels visualize whether each candidate loss prefers unit-length and orthogonal structure at the ideal point. See Table 1 for mathematical definitions.

3.1 Ground truth and ground truther

$\ell_S(X) = D(X^T X, I_D)$ is minimized uniquely by orthonormal matrices and invariant under rotation. However, it is plainly not the only function of X satisfying these properties. Consider that function

$$\ell : \mathbb{R}^{D\times P} \to \mathbb{R}^+ \tag{6}$$

$$X \mapsto \sum_{d=1}^{D} \log g(\sigma_d(X)) \tag{7}$$

where

$$g : \mathbb{R}^+ \to \mathbb{R}^+ \tag{8}$$

$$t \mapsto \frac{t + t^{-1}}{2}. \tag{9}$$

$$g : \mathbb{R}^+ \to \mathbb{R}^+ \tag{10}$$

$$t \mapsto \frac{e^t + e^{t^{-1}}}{2e}. \tag{11}$$

By Proposition 1, we can see that ℓ is uniquely maximized by orthonormal matrices. Moreover, g is convex, and $\ell(X^{-1}) = \ell(X)$ when X is invertible. Figure 2 gives graphs of candidate losses when $D = 1$ and $D = 2$.

Our ground truth objective is therefore the best subset selection problem

$$\arg\min_{S\in\binom{[P]}{d}} \ell(X_{\cdot S}). \tag{12}$$

Loss Function	Definition	Unique min.	Min. at isometry	Prefers unit-length	Prefers orthogonal
Distortion[Kohli et al. (2021)]	$\sigma_{\max}(X)/\sigma_{\min}(X) - 1$	No	Yes	No	Yes
Isometry loss	$\sum_{d=1}^{D} g(\sigma_d(X)) - D$	Yes	Yes	No	Yes
S-divergence to identity	$\log\det\left((X^\top X + I_D)/2\right) - \frac{1}{2}\log\det(X^\top X)$	Yes	Yes	Yes	Yes
Normalized log-det volume	$-\sum_{d=1}^{D} \log \sigma_d\left(n_q^{\times P}(X)\right)$	No	No	Yes	No
Frob. pseudoinverse	$\|n_q^{\times P}(X)^\dagger\|_F - \sqrt{D}$	No	Yes	Yes	Yes
IsometryPursuit	$\|n_q^{\times P}(X)^\dagger\|_{1,2} - D$	Yes	Yes	Yes	Yes

Table 1: Matrix loss functions evaluated for isometry pursuit. We evaluate four key properties: whether the loss has a unique minimum, whether it is minimized specifically at orthonormal matrices (isometries), whether it prefers unit-length vectors over scaled versions, and whether it prefers orthogonal (orthonormal) structure.

Regardless of the convexity of ℓ, brute combinatorial search over $[P]$ is inherently non-convex. This motivates our alternative formulation.

3.2 Normalization

Since basis pursuit methods tend to select longer vectors, selection of orthonormal submatrices requires normalization such that both long and short candidate basis vectors are penalized in the subsequent regression. We introduce the following definition.

Definition 2 (Symmetric normalization) *A function $q : \mathbb{R}^D \to \mathbb{R}^+$ is a symmetric normalization if*

$$\arg\max_{v\in\mathbb{R}^D} \; q(v) = \{v : \|v\|_2 = 1\} \tag{13}$$

$$q(v) = q(\frac{v}{\|v\|_2^2}) \tag{14}$$

$$q(v_1) = q(v_2) \; \forall \; v_1, v_2 \in \mathbb{R}^D : \|v_1\|_2 = \|v_2\|_2. \tag{15}$$

We use such functions to normalize column-vector length in such a way that vectors of length 1 prior to normalization have longest length after normalization and vectors are shrunk proportionately to their deviation from 1. That is, we normalize vectors by

$$n : \mathbb{R}^D \to \mathbb{R}^D \tag{16}$$

$$v \mapsto q(v)\frac{v}{\|v\|_2} \tag{17}$$

and matrices by

$$w : \mathbb{R}^{D\times P} \to \mathbb{R}^{D\times P} \tag{18}$$

$$X_{.p} \mapsto n(X_{.p}) \; \forall \; p \in [P]. \tag{19}$$

We choose q as follows.

$$q : \mathbb{R}^D \to \mathbb{R}^+ \tag{20}$$

$$v \mapsto \frac{2e}{e^{\|v\|_2} + e^{\|v\|_2^{-1}}}. \tag{21}$$

Besides satisfying the conditions in Definition 2, this normalization has some additional nice properties. First, q is convex. Second, it grows asymptotically log-linearly. Third, while $\exp(-|\log t|) = \exp(-\max(t, 1/t))$ is a seemingly natural choice for normalization, it is non smooth, and the LogSumExp (Boyd and Vandenberghe, 2004) replacement of $\max(t, 1/t)$ with $\log(\exp(t)+\exp(1/t))$ simplifies to 21 upon exponentiation. This normalization provides good numerical properties for avoiding issues arising close to 0 and ∞. We plot the effect of this normalization on our example data in Figure 1c.

3.3 Isometry pursuit

Isometry pursuit is the application of multitask basis pursuit to the normalized design matrix $w(X)$ to identify submatrices of X that are as orthonormal as possible. Define the multitask basis pursuit penalty

$$\|\cdot\|_{1,2} : \mathbb{R}^{P\times D} \to \mathbb{R}^+ \tag{22}$$

$$\beta \mapsto \sum_{p=1}^{P} \|\beta_{p.}\|_2. \tag{23}$$

Given a matrix $Y \in \mathbb{R}^{D\times D}$, the multitask basis pursuit solutions are

$$\widehat{\beta}_{MBP}(X, Y) := \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \; : \; Y = X\beta. \tag{24}$$

The multitask basis pursuit solution is often not unique. See Schneider and Tardivel (2022) for more details on uniqueness of basis pursuit solutions, for which the uniqueness condition is less strict than in the multitask case. We therefore define

$$\widehat{\beta}^{\ell_2}_{MLP}(X,Y) := \arg\min \|\beta\|_F : \beta \in \beta_{MBP}(X,Y). \tag{25}$$

By the strong convexity of the ℓ_2 norm, $\widehat{\beta}^{\ell_2}_{MLP}(X,Y)$ is well defined.

This implicit regularization assumption is reasonable. For example, in Tibshirani (2012) the implicit selection of the minimum ℓ_2 norm solution among the lasso solutions was proven for the Least Angle Regression Selection (LARS) algorithm for solving the lasso, while Mishkin and Pilanci (2022) assumes that the group lasso solution is in fact min-norm prior to subsequent theoretical analyses. Empirically, the Splitting Conic Solver type method (which is itself a variant of the Alternating Direction Method of Multilpliers) (O'Donoghue et al., 2013) stably estimates the min-norm solution when initialized at 0, but besides a brief exploration in Section **??** we leave theoretical and experimental proof of this feature of the feature our optimization approach for future work, and instead make the ability of our optimizer to obtain the min-norm solution an assumption of our eventual proposition.

Isometry pursuit is then given by

$$\widehat{\beta}(X) := \widehat{\beta}^{\ell_2}_{MBP}(w(X), I_D) \tag{26}$$

where I_D is the D dimensional identity matrix and recovered functions are the indices of the dictionary elements with non-zero coefficients. That is, they are given by $S(\beta)$ where

$$S : \mathbb{R}^{P\times D} \to \binom{[P]}{D} \tag{27}$$

$$\beta \mapsto \{p \in [P] : \|\beta_{p\cdot}\| > 0\}. \tag{28}$$

IsometryPursuit(Matrix $X \in \mathbb{R}^{D\times P}$)

1: Normalize $X' = w(X)$
2: Optimize $\widehat{\beta} = \widehat{\beta}^{\ell_2}_{MBP}(X', I_D)$
3: **Output** $\widehat{S} = S(\widehat{\beta})$

3.4 Theory

The intuition behind our application of multitask basis pursuit is that submatrices consisting of vectors which are closer to 1 in length and more orthogonal will have smaller loss. A key theoretical assertion is that IsometryPursuit is invariant to choice of basis for X.

Proposition 4 *Let $U \in \mathbb{R}^{D\times D}$ be orthonormal. Then $S(\widehat{\beta}(UX)) = S(\widehat{\beta}(X))$.*

This proposition establishes the basis-independence of Isometry Pursuit and that we may replace I_D in the constraint by any orthonormal $D \times D$ matrix. It is also essential for showing the following main theoretical result.

When a rank D orthonormal column-submatrix $X_{\cdot S}$ exists, the output of Program 26 will contain S.

Proposition 5 *Let $X \in \mathbb{R}^{D\times P}$ have a rank D orthonormal column submatrix $X_{\cdot S}$. Then $S \subseteq \widehat{\beta}(X)$.*

Proofs of these propositions are given in Section 4.2, with corresponding experimental results in Section **??**. The proof mostly relies on the KKT conditions (Hastie et al., 2015) being used to verify that an isometric subset yields a corresponding basis pursuit solution for an appropriate normalized matrix. This is a formalization of our intuition that longer and more orthogonal vectors will minimize loss. We then apply a simple argument to show that the ℓ_2 minimizing solution is the least sparse solution, which means that it contains the isometry solution, should it exist.

3.5 Two-stage isometry pursuit

Proposition 5 suggests the following algorithm, which first uses the convex IsometryPursuit algorithm to prune the P candidate features and then applies brute search upon the reduced feature set. The proposition says that this approach is guaranteed to select an isometric submatrix should one exist. The two-stage approach is standard in the Lasso literature (Meinshausen, 2007; Hesterberg et al., 2008; Koelle et al., 2022), and Proposition 5 justifies it in our application. This method forms our practical isometry estimator, the core advantage of which over purely brute search is a substantially improved computational complexity.

TwoStageIsometryPursuit(Matrix $X \in \mathbb{R}^{D \times P}$)

1: $\widehat{S}_{IP} = \text{IsometryPursuit}(X)$
2: $\widehat{S} = \text{BruteSearch}(X_{\cdot \widehat{S}_{IP}}, \ell)$
3: **Output** $\widehat{S}$

3.6 Non-uniqueness

The isometric solution is optimum among cardinality D solutions.

Proposition 6 $\ell(X_S) = D, |S| = D \Leftrightarrow X_S X_S^T = I_D$

However, there exist Isometry Pursuit solutions of minimum norm that are not isometries. For example, *Mercedes-Benz frame* matrix X achieves minimium norm.

Example 1 *Let* $D = 2, P = 3$. *Let* $X_S = \begin{bmatrix} 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}, \beta = \begin{bmatrix} 0 & \frac{2}{3} \\ \frac{1}{\sqrt{3}} & -\frac{1}{3} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{3} \end{bmatrix}$ *Then* $X\beta = I_D$ *and* $\|\beta\|_{1,2} = \frac{2}{3} + \frac{2}{3} + \frac{2}{3} = 2 = D$. *Since the identity matrix is a solution regardless of* X_{-S}, *this is a non-isometric minimum loss solution.*

The upshot of this example is that IsometryPursuit can select solutions that are relatively non-isometric. The Mercedes-Benz frame is an example of a tight frame - a frame whose non-zero singular values are equal. While there are non-tight frames that have minimizing solutions, we do not here resolve whether there exist tight frames with unit norm columns that have minimum norm solutions.

3.7 Computation complexity

IsometryPursuit is more computationally efficient than BruteSearch search for the minimizer of ℓ. Interestingly, their asymptotic per-iteration complexities are the same.

Proposition 7 *The computational complexity* C_ℓ *of 6 is* $O(D^2 P)$.

The above proposition follows immediately from the computational complexity of singular value decomposition (Liang et al., 2016). The per-iteration complexity of IsometryPursuit is also limited by the computational complexity of singular value decomposition, albeit in the more intricate manner described in Section 4.2.5.

Proposition 8 *The computational complexity of each step of Isometry Pursuit is* $O(D^2 P)$.

The difference therefore lies in the relation of the number of iterations required to achieve a given error tolerance with IsometryPursuit and BruteSearch. To optimize, we use the SCS interior point solver (O'Donoghue et al., 2016) from CVXPY (Diamond and Boyd, 2016; Agrawal et al., 2018), which is able to push sparse values arbitrarily close to 0 (CVXPY Developers). SCS (aka ADMM) has asymptotic (in error) accuracy $\epsilon < O(\frac{1}{\mathcal{S}})$, where $\mathcal{S}$ is the number of iterations (He and Yuan, 2012). This means that the number of iterations

required to achieve a given error tolerance scales inversely with the desired level of accuracy. We examine the effect of iteration number on the convergence to a sparse solution in Section **??**.

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4 Supplement

This section contains algorithms, proofs, and experiments in support of the main text.

4.1 Algorithms

We give definitions of the brute and greedy algorithms for the combinatorial problem studied in this paper. The brute force algorithm is computationally intractable for all but the smallest problems, but always finds the global minima.

BruteSearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f)

1: **for** each combination $S \subseteq \{1, 2, \ldots, P\}$ with $|S| = D$ **do**
2: Evaluate $f(X_{.S})$
3: **end for**
4: **Output** the combination S^* that minimizes $f(X_{.S})$

Greedy algorithms are computationally expedient but can get stuck in local optima (Cormen et al., 2009; Russell and Norvig, 2009), even with randomized restarts (Dick et al., 2014).

GreedySearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f, selected set $S = \emptyset$, current size $d = 0$)

1: **if** $d = D$ **then**
2: **Return** S
3: **else**
4: **Initialize** $S_{\text{best}} = S$
5: **Initialize** $f_{\text{best}} = \infty$
6: **for** each $p \in \{1, 2, \ldots, P\} \setminus S$ **do**
7: **Evaluate** $f(X_{.(S \cup \{p\})})$
8: **if** $f(X_{.(S \cup \{p\})}) < f_{\text{best}}$ **then**
9: **Update** $S_{\text{best}} = S \cup \{p\}$
10: **Update** $f_{\text{best}} = f(\mathcal{X}_{.(S \cup \{p\})})$
11: **end if**
12: **end for**
13: **Return** GreedySearch(X, f, S_{best}, $d + 1$)
14: **end if**

4.2 Proofs

4.2.1 Proof of Proposition 3

Proof: First,

$$\frac{1}{2} \log \det(X^T X) = \frac{1}{2} \log \prod_{d=1}^{D} \sigma_d(X)^2 \tag{29}$$

$$= \log \prod_{d=1}^{D} \sigma_d(X) \tag{30}$$

$$= \sum_{d=1}^{D} \log \sigma_d(X). \tag{31}$$

Secondly, I_D and X^TX are diagonalizable in the same basis, so

$$\log\det(\frac{X^TX+I_D}{2}) = \log\prod_{d=1}^{D}\frac{\sigma_d(X)^2+1}{2} \tag{32}$$

$$= \sum_{d=1}^{D}\log\frac{\sigma_d(X)^2+1}{2}. \tag{33}$$

Then,

$$D(X^TX, I_D)^2 = \sum_{d=1}^{D}(\log\frac{\sigma_d(X)^2+1}{2} - \log\sigma_d(X)) \tag{34}$$

$$= \sum_{d=1}^{D}\log\frac{\sigma_d(X)^2+1}{2\sigma_d(X)} \tag{35}$$

$$= \sum_{d=1}^{D}\log\frac{\sigma_d(X)+\sigma_d(X)^{-1}}{2}. \tag{36}$$

□

4.2.2 Proof of Proposition 4

In this proof we first show that the penalty $\|\beta\|_{1,2}$ is unchanged by unitary transformation of β.

Proposition 9 *Let $U \in \mathbb{R}^{D\times D}$ be unitary. Then $\|\beta\|_{1,2} = \|\beta U\|$.*

Proof:

$$\|\beta U\|_{1,2} = \sum_{p=1}^{P}\|\beta_{p.}U\| \tag{37}$$

$$= \sum_{p=1}^{P}\|\beta_{p.}\| \tag{38}$$

$$= \|\beta\|_{1,2} \tag{39}$$

□

We then show that this implies that the resultant loss is unchanged by unitary transformation of X.

Proposition 10 *Let $U \in \mathbb{R}^{D\times D}$ be unitary. Then $\widehat{\beta}(UX) = \widehat{\beta}(X)U$.*

Proof:

$$\widehat{\beta}(UX) = \arg\min_{\beta\in\mathbb{R}^{P\times D}}\|\beta\|_{1,2} \ : \ I_D = UX\beta \tag{40}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}}\|\beta\|_{1,2} \ : \ U^{-1}U = U^{-1}UX\beta U \tag{41}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}}\|\beta\|_{1,2} \ : \ I_D = X\beta U \tag{42}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}}\|\beta U\|_{1,2} \ : \ I_D = X\beta U \tag{43}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}}\|\beta\|_{1,2} \ : \ I_D = X\beta. \tag{44}$$

□

4.2.3 Proof of Proposition 5

In this proof we first show that the orthonormal submatrix $X_{\cdot S}$ is contained within the set $\widehat{\beta}_{MBP}(w(X), I_D)$.

Proposition 11 *Let $X_{\cdot S}$ be a rank D orthonormal submatrix of X. Then $S \in S(\beta) : \beta \in \widehat{\beta}_{MBP}(w(X), I_D))$.*

Proof: By Proposition 4 without loss of generality, let $X_{\cdot S} = I_D$. Also, without loss of generality let $S = \{1 \dots D\}$. We show that $\beta = [I_D \; 0]$ satisfies the KKT conditions for $X = [I_D \; X_{-S}]$.

Recall that the KKT conditions for a convex optimization problem are a set of conditions on an element of the domain of the optimization algorithm that, if satisfied, certify that the element is a solution. To write out the KKT conditions, we define the Lagrangian

$$\mathcal{L}(X, \beta, \nu) = \|\beta\|_{1,2} + \nu^T (I_D - w(X)\beta) \tag{45}$$

where $\nu \in \mathbb{R}^{D \times D}$ is the dual variable.

The KKT conditions are then

- Primal feasibility: $w(X)\beta = I_D$
- Stationarity: there exists a dual variable ν such that $0 \in \partial \|\beta\|_{1,2} - w(X)^T \nu$ where ∂ is the subdifferential operator.

where dual feasibility and complementary slackness are ignorable by virtue of the absence of inequality constraints.

First note that in our case, primal feasibility is satisfied by X being rank D and w not effecting the rank of the transformed matrix since it only rescales length and not direction.

For stationarity, recall that

$$\|\beta\|_{1,2} = \sum_{p=1}^{P} \|\beta_{p\cdot}\|_2. \tag{46}$$

Then, by the definition of subdifferential

$$\partial \|\beta\|_{1,2} = \begin{bmatrix} I_D \\ v_{d+1} \\ \dots \\ v_P \end{bmatrix} \tag{47}$$

where $v_p \in \mathbb{R}^D$ can be any vector satisfying $\|v_p\|_2 \leq 1$.

We therefore must show that there exists a ν that satisfies

$$\begin{bmatrix} I_D \\ w(X_{-S})^T \end{bmatrix} \nu = \begin{bmatrix} I_D \\ V_{-S} \end{bmatrix}. \tag{48}$$

At this point we can see that in fact I_D is an appropriate choice of ν since normalization by w leads to vectors satisfying $\|w(X)\|_2 \leq 1$.

Since for this β, $S(\beta) = S$, we have shown that S is the support of a solution to $\beta_{MBP}(w(X), I_D)$. □

The next part of the proof handles the presence of non-unique solutions.

Proposition 12 $S(\widehat{\beta}(X)) = \cup \, S(\beta) \; : \; \beta \in \beta_{MBP}(w(X), I_D)$

Proof: The proposition follows from geometric properties of the solution set. We ignore the case for singleton solutions, for which the proposition is self-evident. Recall that $\widehat{\beta}(X) := \widehat{\beta}^{\ell_2}_{MBP}(w(X), I_D) = \arg\min_{\beta \in \mathbb{R}^{P \times D}} \|\beta\|_F : \beta \in \widehat{\beta}_{MBP}(X, I_D)$. By definition of

$\widehat{\beta}_{MBP}(X, I_D)$, $\|\beta_{p\cdot}\|_2 : \beta \in \widehat{\beta}_{MBP}(X, I_D)$ is an affine linear subspace of $\mathbb{R}^P$. The projection of the vector of minimum ℓ_2 distance from the origin to a affine space intersects that space perpendicularly. However, for all $\|\beta_{\cdot p}\|_2$ such that β is on the intersection of this affine space with the coordinate hyperplanes formed by fixing any choices of $\|\beta_{\cdot p}\|_2 = 0$, this vector doesn't intersect this space perpendicularly. Thus, $\widehat{\beta}(X)$ is not on the intersection of the solution polytope $\widehat{\beta}_{MBP}(w(X), I_D)$ with the coordinate hyperplanes. It is therefore maximally non-sparse with respect to any of the minimizing solutions. □

4.2.4 Proof of Proposition 6

X_S admits a QR decomposition $X_S = QR$ where Q is orthogonal and R is upper triangular. By Proposition 4, we can assume without loss of generality that $Q = I_D$. That is, X_S is upper triangular. Thus, $\beta = X_S^{-1}$ is upper triangular with diagonal elements $\beta_{dd} = R_{dd}^{-1}$. $\|\beta_{d\cdot}\|_2 \geq \beta_{dd}$, with equality only occurring when the off-diagonal elements are 0. Therefore, $\|\beta_{1,2}\| \geq D$ with equality occurring only when $\beta = I_D$.

4.2.5 Proof of Proposition 8

Proof: The augmented split Lagrangian minimized by ADMM is

$$\mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z) = \|z\|_{1,2} + \text{tr}(\nu^T(I_D - w(X)\beta)) + \rho\|I_D - w(X)\beta\|_F^2 + \gamma\|\beta - z\|_F^2 \tag{49}$$

This minimization problem is

$$\arg \min_{\beta, z \in P \times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z) \tag{50}$$

For a more detailed explanation of the reasons to prefer optimizing the split augmented Lagrangian over direct optimization of Equation 22 see Boyd (2010).

The three main subroutines (54 - 58) called by ADMM determine the per iteration complexity. The update rules of the ADMM algorithm used to optimize the augmented split Lagrangian are

$$\beta^{k+1} = \arg \min_{\beta \in P \times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z^k) \tag{51}$$

$$z^{k+1} = \arg \min_{z \in P \times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta^{k+1}, z) \tag{52}$$

$$\nu^{k+1} = \nu^k + \rho(I_D - w(X)\beta^{k+1}) \tag{53}$$

We will analyze the computational complexity of each of these steps by rewriting the update rules according to the specifics of the isometry pursuit problem.

The beta update Equation 54 projects the solution onto the feasible set.

$$\beta^{k+1} = \arg \min_{\beta \in P \times D} \text{tr}({\nu^k}^T(I_D - w(X)\beta)) + \rho\|I_D - w(X)\beta\|_2^2 + \gamma\|\beta - z^k\|_2^2 \tag{54}$$

$$= (\rho\, w(X)^T w(X) + \gamma I_D)^{-1}(\rho\, w(X)^T I_D + \gamma z^k + w(X)^T \nu^k) \tag{55}$$

The computational complexity of this computation is dominated by the inverse - that is, $O(D^2P)$.

Subsequently, the z update Equation 56 is soft-thresholding operation onto a contracted level set $\|\|_{1,2}$ ball.

$$z^{k+1} = \arg \min_{z \in P \times D} \|z\|_{1,2} + \gamma\|z - \beta^{k+1}\|_2^2 \tag{56}$$

$$= \begin{cases} (1 - \frac{1}{\gamma\|\beta_{j\cdot}^{k+1}\|_2})(\beta_{j\cdot}^{k+1}) \text{ if } \|\beta_{j\cdot}^{k+1}\|_2 > \frac{1}{\gamma} \\ 0 \text{ otherwise} \end{cases} \tag{57}$$

The computational complexity of this operation is $O(PD)$.

Finally, the ν update Equation 58 increases the squeezing of the z and β variables towards to each other.

$$\nu^{k+1} = \nu^k + \rho(I_D - w(X)\beta^{k+1}) \tag{58}$$

In this step, the dual variable ν that controls the penalty accrued by the splitting is updated in the direction of the constraint violation. The computational complexity of this operation is dominated by that of the matrix multiplication $w(X)\beta^{k+1}$. That is, it is $O(D^2P)$. □

4.3 Partial derivatives of candidate losses

We summarize the column-wise derivatives of the losses in Table 1. Throughout this section let $X = [x_1, \dots, x_D] \in \mathbb{R}^{D\times D}$ be full rank. For a scalar loss $L(X)$, write

$$\nabla_X L = \left[\frac{\partial L}{\partial x_1}, \dots, \frac{\partial L}{\partial x_D}\right]. \tag{59}$$

All formulas are stated on smooth strata: rank deficient matrices, repeated singular values, and zero rows of pseudoinverse factors should be handled using subgradients.

We will use two common identities. First, if $X = U\,\mathrm{diag}(\sigma)V^\top$ is a singular value decomposition and $L(X) = \sum_{d=1}^{D} \phi(\sigma_d(X))$, then

$$\nabla_X L = U\,\mathrm{diag}(\phi'(\sigma_1), \dots, \phi'(\sigma_D))V^\top. \tag{60}$$

Second, for normalized losses let $N = n_q^{\times D}(X)$. If $M = \nabla_N L$, then the column-wise chain rule is

$$\frac{\partial L}{\partial x_p} = \mathrm{D}n_q(x_p)^\top M_{\cdot p}. \tag{61}$$

For $r = \|x\|_2$, write $n_q(x) = s(r)x$ with $s(r) = q(r)/r$. Then

$$\mathrm{D}n_q(x) = s(r)I_D + \frac{s'(r)}{r}xx^\top, \tag{62}$$

$$s'(r) = \frac{q'(r)}{r} - \frac{q(r)}{r^2}, \tag{63}$$

$$q'(r) = -\frac{2e(e^r - r^{-2}e^{1/r})}{(e^r + e^{1/r})^2}. \tag{64}$$

Since $\mathrm{D}n_q(x)$ is symmetric, the transpose in this chain rule may be omitted.

Distortion. Let $\sigma_1 = \sigma_{\max}(X)$ and $\sigma_D = \sigma_{\min}(X)$ with corresponding singular vectors u_1, v_1 and u_D, v_D. For

$$L_{\mathrm{dist}}(X) = \frac{\sigma_1}{\sigma_D} - 1, \tag{65}$$

The spectral gradient identity gives

$$\nabla_X L_{\mathrm{dist}} = \frac{u_1 v_1^\top}{\sigma_D} - \frac{\sigma_1}{\sigma_D^2} u_D v_D^\top. \tag{66}$$

The derivative is insensitive to uniform rescaling, which is the sense in which this loss does not prefer unit length.

Isometry loss. For

$$L_{\mathrm{iso}}(X) = \sum_{d=1}^{D} g(\sigma_d(X)) - D, \tag{67}$$

we have

$$g'(t) = \frac{e^t - t^{-2}e^{1/t}}{2e}. \tag{68}$$

Thus

$$\nabla_X L_{\mathrm{iso}} = U\,\mathrm{diag}(g'(\sigma_1), \dots, g'(\sigma_D))V^\top. \tag{69}$$

S-divergence to identity. Let $G = X^\top X$ and

$$L_{\mathrm{sdiv}}(X) = \log\det\left(\frac{G + I_D}{2}\right) - \frac{1}{2}\log\det(G). \tag{70}$$

The constant $\log\det(1/2)$ does not contribute to the derivative. Using $d\log\det A = \mathrm{tr}(A^{-1}dA)$ and $dG = dX^\top X + X^\top dX$,

$$\nabla_X L_{\mathrm{sdiv}} = 2X(G + I_D)^{-1} - XG^{-1}. \tag{71}$$

Normalized log-det volume. Let $N = n_q^{\times D}(X)$ and

$$L_{\mathrm{nld}}(X) = -\sum_{d=1}^{D} \log \sigma_d(N). \tag{72}$$

For square full-rank N, this is $-\log|\det N|$, so

$$\nabla_N L_{\mathrm{nld}} = -N^{-\top}. \tag{73}$$

Therefore, by the column-wise chain rule,

$$\frac{\partial L_{\mathrm{nld}}}{\partial x_p} = -\mathrm{D}n_q(x_p)^\top (N^{-\top})_{\cdot p}. \tag{74}$$

Frobenius pseudoinverse. Let $N = n_q^{\times D}(X)$ and $B = N^{-1}$. For

$$L_{\mathrm{frob}}(X) = \|N^{-1}\|_F - \sqrt{D}, \tag{75}$$

the differential identity $dB = -B(dN)B$ gives

$$\nabla_N L_{\mathrm{frob}} = -\frac{N^{-\top} N^{-1} N^{-\top}}{\|N^{-1}\|_F}. \tag{76}$$

Thus

$$\frac{\partial L_{\mathrm{frob}}}{\partial x_p} = \mathrm{D}n_q(x_p)^\top \left(-\frac{N^{-\top} N^{-1} N^{-\top}}{\|N^{-1}\|_F} \right)_{\cdot p}. \tag{77}$$

Isometry pursuit proxy. Let $N = n_q^{\times D}(X)$ and $B = N^{-1}$. The differentiable proxy for the IsometryPursuit table row is

$$L_{\mathrm{ip}}(X) = \|N^{-1}\|_{1,2} - D. \tag{78}$$

Define $R \in \mathbb{R}^{D \times D}$ by

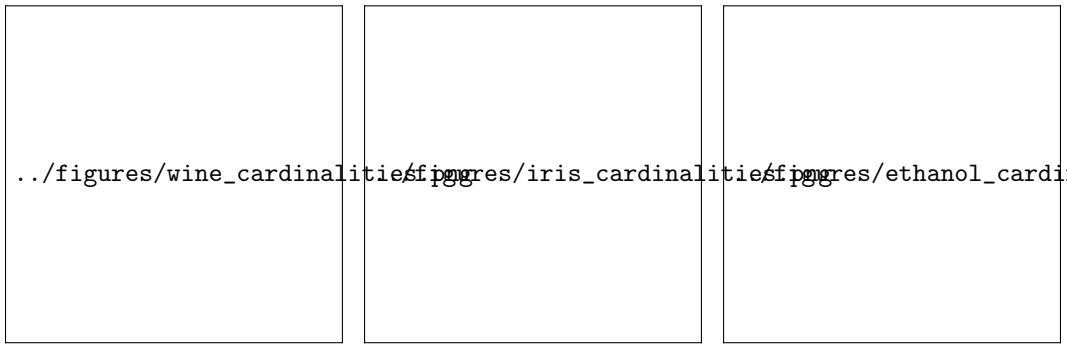
$$R_{p\cdot} = \frac{B_{p\cdot}}{\|B_{p\cdot}\|_2}. \tag{79}$$

Then $R \in \partial \|B\|_{1,2}$ on the smooth stratum where every row of B is nonzero, and

$$\nabla_N L_{\mathrm{ip}} = -B^\top R B^\top. \tag{80}$$

The column-wise gradient is therefore

$$\frac{\partial L_{\mathrm{ip}}}{\partial x_p} = \mathrm{D}n_q(x_p)^\top (-B^\top R B^\top)_{\cdot p}. \tag{81}$$



(a) Wine (b) Iris (c) Ethanol

Figure 3: Support cardinalities for each dataset after IsometryPursuit.

4.4 Support cardinalities

Figure 3 plots the distribution of $|\widehat{S}_{IP}|$ from Table **??** in order to contextualize the reported means. While typically $|\widehat{S}_{IP}| << P$, there are cases for Ethanol where this is not the case that drive up the means. Further investigation into the extreme failure of isometry pursuit to sparsify in the the minor chunk of the bimodal Ethanol support cardinality distribution is warranted.

4.5 Timing

While wall-time of algorithms is a non-theoretical quantity that depends on implementation details, it provides valuable context for practitioners. We therefore report the following runtimes on a 2021 Macbook Pro. The particularly high variance for brute force search in the second step of TwoStageIsometryPursuit is likely due to the large cardinalities reported in Figure 3.

Name	IP	2nd stage	Greedy
Iris	1.24 ± 0.02	0.00 ± 0.00	0.02 ± 0.00
Wine	2.32 ± 0.17	0.13 ± 0.12	0.03 ± 0.00
Ethanol	8.38 ± 0.57	0.55 ± 1.08	0.07 ± 0.01

Table 2: Algorithm runtimes in seconds across replicates.

Example 2

$$X_S = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} & \frac{-1}{2} \\ \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & 0 & 0 & & \\ 0 & 0 & \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 00 & 0 & \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & \end{bmatrix}. \tag{82}$$

Proof: Let

By Given the symmetry of the second and third columns around the y axis, the loss will not change regardless of whether we upweight one or downweight the other, these changes necessarily being in conjunction. We therefore assume that their loadings with respect to the second column of the target matrix (the D dimensional identity) are the same. Clearly, they will both then have value of $\frac{1}{\sqrt{3}}$. The question then becomes, what should the values of the first column in the loading matrix β be?

We can answer this question with a simple differential equation. Let b denote the constraint violation $1 - X_{1.}\beta_{.1}$.

Then

$$\frac{\partial b}{\partial \beta_{11}} = \frac{\partial b}{\partial \beta_{21}} \tag{83}$$

$$= \frac{1}{2} \tag{84}$$

On the other hand, due to the value of $\frac{1}{\sqrt{3}}$ already assumed for the second loading of the second and third features,

$$\frac{\partial l}{\partial \beta_{11}} = \frac{\partial l}{\partial \beta_{21}} \tag{85}$$

$$= \frac{\partial}{\partial x}\sqrt{\frac{1}{3} + x^2} \tag{86}$$

$$= \frac{1}{2}(\frac{1}{3} + x^2)^{-\frac{1}{2}}(2x) \tag{87}$$

$$= \frac{x}{\sqrt{\frac{1}{3} + x^2}}. \tag{88}$$

This means that the tradeoff

$$\frac{\partial l}{\partial b} = \frac{1}{2x}\sqrt{\frac{1}{3} + x^2}. \tag{89}$$

In contrast, $\frac{\partial l}{\partial \beta_{11}} = 1$ and similarly for its constraint violation. Solving for the equilibrium, we get

$$\frac{1}{2x}\sqrt{\frac{1}{3} + x^2} = 1 \tag{90}$$

$$\frac{1}{3} + x^2 = 4x^2 \tag{91}$$

$$\frac{1}{3} = 3x^2 \tag{92}$$

$$x = \frac{1}{\sqrt{3}}. \tag{93}$$

This leads to a value for l of

$$\sqrt{\frac{1}{3} + \frac{1}{9}} + \sqrt{\frac{1}{3} + \frac{1}{9}} + \frac{2}{3} \tag{94}$$

$$= \sqrt{\frac{4}{9}} + \sqrt{\frac{4}{9}} + \frac{2}{3} \tag{95}$$

$$= \frac{2}{3} + \frac{2}{3} + \frac{2}{3} \tag{96}$$

□